Are Player Salaries Accurate to their Performance?

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Stat 441 - Project 1

Background

- In 2026, the New York Mets spent more than $350 million on players and finished last in their division.
- Spending does not guarantee winning. What matters is whether teams pay for the right players.

National League East

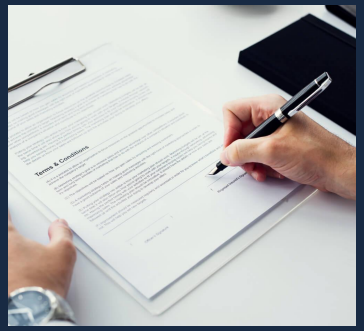
Team	2026 Payroll	W	L	Pct
Braves	$263 M	94	68	.580
Phillies	$292 M	88	74	.543
Marlins	$83 M	80	82	.494
Nationals	$101 M	77	85	.475
Mets	$350 M	74	88	.457

Driving Question:

Can WAR explain MLB player salaries, and which players provide more or less value than their contracts would predict?

Why it Matters

- Teams have finite payroll and luxury tax thresholds, so misallocated dollars cost wins

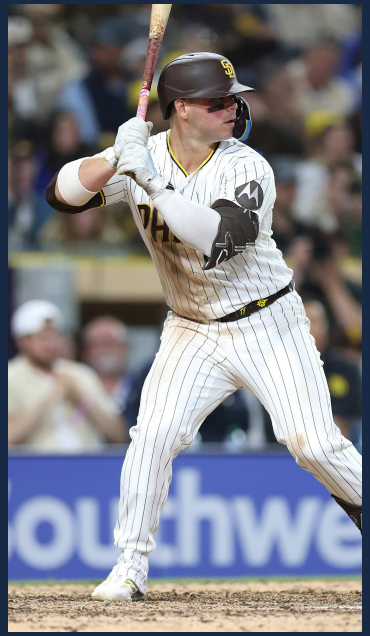


- **Useful for**:
  - Front offices (contract decisions)
  - Agents (negotiations)
  - Fans (evaluating GM decisions)

- **WAR (Wins Above Replacement)** is an advanced sabermetric statistic in baseball that estimates the total number of wins a player contributes to their team above what a readily available, low-cost "replacement-level" player (someone who is under the top 390 in plate-appearances) would provide

Limitations

- Salaries are set before the season – contracts are forecasts, not rewards
- Arbitration and rookie contracts distort the market
- Salary does not explain performance for arb/pre-arb because they are not open market players yet
- WAR on its own only takes performance into account, not performance compared to salary



The Question

- Created custom 2026 WAR from Statcast (combined hitting and fielding)
- Free agents only, $ per 1.0 WAR
- Compare everyone else with a salary to that price
- Value WAR = on-field WAR minus the wins their salary bought

Analysis Overview: Data and Cleaning

- Used the “sabRmetrics” package in RStudio to download all of the 2026 MLB statcast play-by-play data
- 2026, regular season only
- Filtered the data to only hold the plate appearance ending pitches (strikeout, walk, HBP, single, double, etc.),
- Rolled that up to one row per batter
- Joined 2026 salaries on batter ID
- Players in the top 390 in PA selected, under top 390 is considered replacement level

Batting

- Each PA has a run-expectancy change (delta_run_exp)
- Sum that up for the whole season, -> offensive runs
- Subtract league average runs for the same PA, -> batting RAA
- Hitting better or worse than average

Fielding

- Balls in play, who fielded it, hit or out
- Score with xBA vs actual, hit prevention given how hard the ball was hit
- Credit the fielder and multiply xBA - hit by ⅓ to convert to runs
- Infielders are typically credited more in this system than outfielders because hard contact hit at them has a high xBA

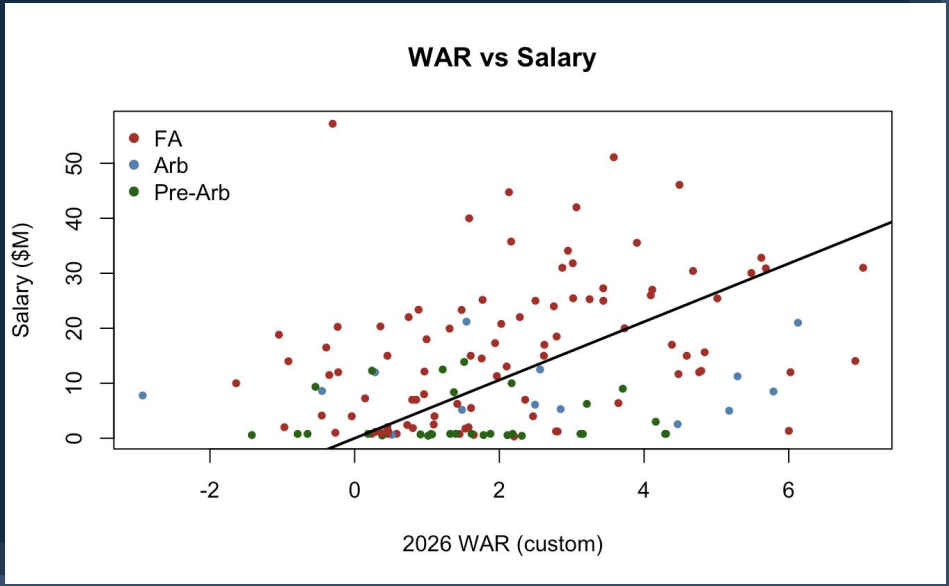
Incorporating Salary

- Regulars are considered top 390 in plate appearances
- Everyone else is considered replacement level
- WAR = (RAA - replacement) / 10
- Free agents only, through 0 = $5.3 million per 1.0 WAR
- Salary in win units = salary / 5.3
- Value WAR = WAR - salary / 5.3
- Surplus $ = expected pay - actual pay

Does WAR explain salary?

- No, 2026 WAR does not explain 2026 Free Agent salary
- Line helped determine the price of 1.0 WAR
- 1.0 WAR = $5.3 million
- Used this to incorporate performance compared to salary in our own WAR stat

- Free Agent correlation = 0.36
- R^2 = 0.13



Bargain vs Overpaid Position Players

- Largest Overpay:
  - Kyle Tucker (LAD)
    - $57.2M
    - Replacement-Level Production
- Biggest Bargain:
  - Ty France (SD)
    - $1.4M
    - Valued at $31M

	Player	Team	WAR	Salary ($M)	Value ($M)	Surplus ($M)	WAR Over Contract
1	Tucker, Kyle	LAD	−0.3	57.2	0.0	−57.2	−11.1
2	Bellinger, Cody	NYY	2.1	44.8	11.3	−33.4	−6.3
3	Soto, Juan	NYM	3.6	51.1	19.0	−32.1	−6.1
4	Judge, Aaron	NYY	1.6	40.0	8.4	−31.6	−6.0
5	Bichette, Bo	NYM	3.1	42.0	16.2	−25.8	−4.9
6	Guerrero Jr., Vladimir	TOR	2.2	35.8	11.4	−24.3	−4.6
7	Ohtani, Shohei	LAD	4.5	46.1	23.8	−22.3	−4.2
8	Nimmo, Brandon	TEX	−0.2	20.2	0.0	−20.2	−4.1
9	Lee, Jung Hoo	SF	−1.0	18.8	0.0	−18.8	−4.6
10	Yelich, Christian	MIL	0.9	23.4	4.7	−18.7	−3.5

	Player	Team	WAR	Salary ($M)	Value ($M)	Surplus ($M)	WAR Over Contract
1	France, Ty	SD	6.0	1.4	31.8	30.4	5.7
2	Contreras, Willson	BOS	6.9	14.1	36.6	22.6	4.3
3	Arraez, Luis	SF/PHI	6.0	12.0	31.9	19.9	3.8
4	Lowe, Nathaniel	CIN/CLE	2.8	1.2	14.9	13.6	2.6
5	Vargas, Ildemaro	AZ	2.8	1.2	14.7	13.5	2.5
6	Díaz, Yandy	TB	4.8	12.0	25.2	13.2	2.5
7	Muncy, Max	LAD	4.8	12.2	25.4	13.1	2.5
8	Castro, Willi	COL	3.6	6.4	19.3	12.9	2.4
9	Hoerner, Nico	CHC	4.5	11.7	23.7	12.0	2.3
10	Walton, Donovan	ATH/LAA	2.2	0.3	11.7	11.3	2.1

Conclusion

Approach

- WAR → dollar value vs. actual salary

Findings

  - Star contracts are the biggest overpays
  - Top bargains: productive players on cheap deals
  - Pay tracks reputation more than current production

Further Model Development:

- Expanding to all MLB Position Players, not just Free Agents
- Tightening the WAR calculation
  - Update Fielding
  - Account for Position Difficulty
- Introduce additional seasons and determine if their average WAR is deserving of their contract